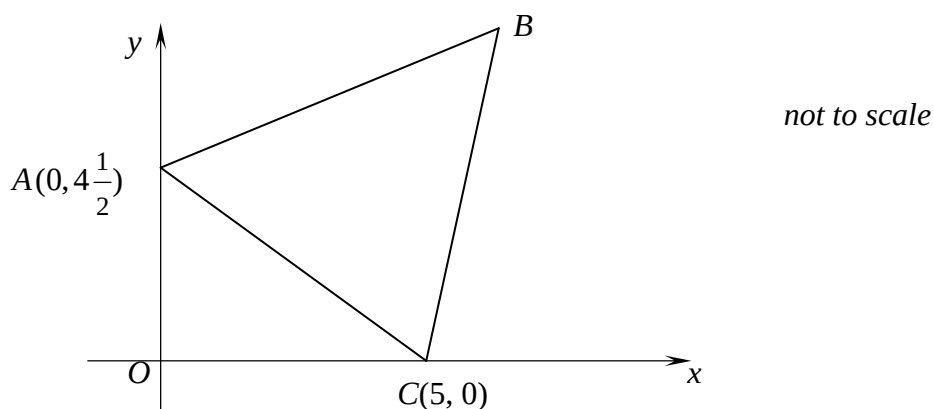


**Question 1.****Marks**

- (a) Evaluate  $\frac{e^3 - 2 \cdot 1^2}{\sqrt{3 \cdot 14 + 2 \cdot 1}}$  correct to 2 significant figures. **1**
- (b) Given that  $\tan \theta = \frac{7}{8}$  and  $\cos \theta < 0$ , find the exact value of  $\operatorname{cosec} \theta$ . **2**
- (c) Solve for  $x$ :  $|4x - 15| \leq 3$ . **2**
- (d) State the period and the amplitude for the graph of  $3y = \sin\left(2x - \frac{\pi}{4}\right)$ . **2**
- (e) Paint at the local hardware store is sold at a profit of 30% on the cost price. If a can of paint is sold for \$67.60, find the cost price. **2**
- (f) Solve for  $\alpha$ :  $\tan \alpha = -0.5$ , where  $0 < \alpha < \pi$ , correct to 2 decimal places. **1**
- (g) Two fair dice are rolled at random. Find the probability that the two numbers are the two digits of a perfect square? **2**

**Question 2.****[START A NEW PAGE]**

The lines  $AB$  and  $CB$  have equations:  $x - 2y + 9 = 0$  and  $4x - y - 20 = 0$  respectively.

- (a) Show that the equation of the line  $AC$  is  $9x + 10y - 45 = 0$ . **2**
- (b) Calculate the exact distance  $AC$ . **1**
- (c) Find the coordinates for point  $B$ . **2**
- (d) Find the angle of inclination of the line through  $A$  and  $B$  (to nearest degree). **2**
- (e) Calculate the shortest distance from point  $B$  to the line  $AC$ . Hence find the area of triangle  $ABC$ . **2**
- (f) Determine the inequalities that define the area bounded by  $\triangle ABC$ . **3**

|                    |                           |              |
|--------------------|---------------------------|--------------|
| <b>Question 3.</b> | <b>[START A NEW PAGE]</b> | <b>Marks</b> |
|--------------------|---------------------------|--------------|

- |     |                                                                                     |                                                        |   |
|-----|-------------------------------------------------------------------------------------|--------------------------------------------------------|---|
| (a) | Find $\frac{dy}{dx}$ for                                                            |                                                        |   |
|     | (i)                                                                                 | $y = (1 + \ln x)^2$ .                                  | 2 |
|     | (ii)                                                                                | $y = \frac{\sin x}{e^{3x} + 1}$ .                      | 2 |
| (b) | (i)                                                                                 | Find $\int (5x - 1)^3 dx$ .                            | 2 |
|     | (ii)                                                                                | Find the value: $\int_0^{\pi} \sec^2 \frac{x}{4} dx$ . | 2 |
| (c) | Given that $\alpha$ and $\beta$ are the roots of the equation $2x^2 - 6x - 7 = 0$ , |                                                        |   |
|     | (i)                                                                                 | Find $\alpha + \beta$ .                                | 1 |
|     | (ii)                                                                                | Find $\alpha^2 + \beta^2$ .                            | 2 |
|     | (iii)                                                                               | Find $\alpha^2 - 3\alpha$ .                            | 1 |

|                    |                           |
|--------------------|---------------------------|
| <b>Question 4.</b> | <b>[START A NEW PAGE]</b> |
|--------------------|---------------------------|

- |     |                                                                                                                                                                      |                                                                |   |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|---|
| (a) | The good ship Lollypop sails from port A 60 nautical miles due west to port B. It then sails a distance of 50 nautical miles on a bearing of $210^\circ$ T to port C |                                                                |   |
|     | (i)                                                                                                                                                                  | Draw a diagram to illustrate this information.                 | 1 |
|     | (ii)                                                                                                                                                                 | Calculate the distance of C from A (to nearest nautical mile). | 2 |
|     | (iii)                                                                                                                                                                | Calculate the bearing of port C from port A.                   | 2 |
| (b) | Evaluate:                                                                                                                                                            | $\lim_{x \rightarrow 16} \frac{x - 16}{\sqrt{x} - 4}$ .        | 2 |
| (c) | Find the equation of the normal to the curve $y = 1 + \ln 2x$ at the point $\left(\frac{e}{2}, 2\right)$ .                                                           |                                                                | 3 |
| (d) | Find the radius and the centre of the circle whose equation is:                                                                                                      |                                                                | 2 |

$$4x^2 - 4x + 4y^2 + 24y + 21 = 0.$$

**Question 5.**

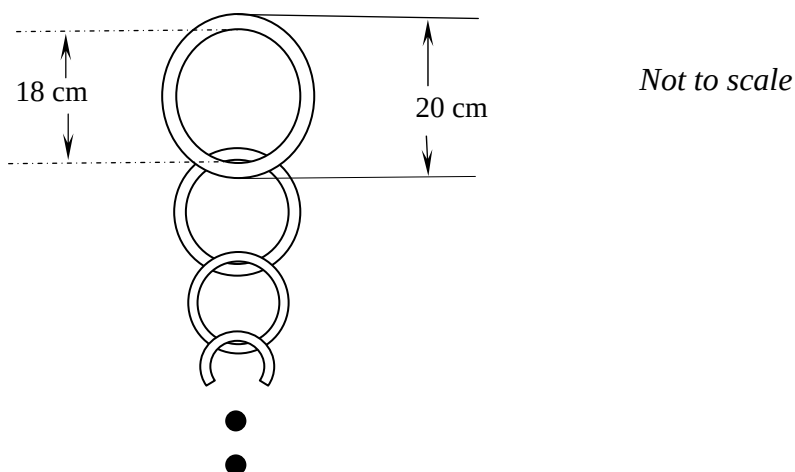
**[START A NEW PAGE]**

**Marks**

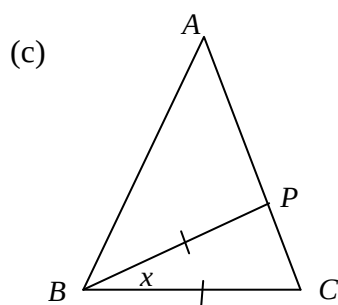
- (a) 1 000 tickets are sold in a raffle. First prize is \$1 000, second prize is \$500 and third prize of \$200.  
The prize winning tickets are drawn consecutively without replacement where the first ticket wins first prize.

- (i) Find the probability that a person buying one ticket in the raffle wins:
- |                     |          |
|---------------------|----------|
| (α) First prize.    | <b>1</b> |
| (β) at least \$500. | <b>1</b> |
| (γ) No prizes.      | <b>1</b> |
- (ii) A person buying two tickets in the raffle wins at least \$500. **2**

- (b) A number of linked rings, each 1 cm thick, are hung from a nail on a wall. The top ring has an outside diameter of 20 cm as shown in the diagram. The outside diameter of each of the other rings is 1 cm less than that of the ring above it. The bottom (last) ring has an outside diameter of 3 cm.



- (i) **Copy** the diagram and explain why the top of the second ring is 17 cm above the top of the third ring. **1**
- (ii) Hence calculate the distance from the top of the top ring to the bottom of the bottom ring. **3**



*Not to scale*

Given  $\triangle ABC$  is an isosceles triangle with  $AB = AC$ . **3**  
 $P$  lies on  $AC$  such that  $\angle ABP = 3\angle PBC$  and  $BP = BC$ .

**Copy** the diagram into your writing booklet and by letting  $\angle CBP = x$ , or otherwise, find angle  $\angle CBP$  expressing it in radians.

**Question 6.****[START A NEW PAGE]****Marks**

(a) Show that  $\frac{1 + \tan^2 A}{\operatorname{cosec}^2 A} = \tan^2 A.$  **2**

(b) Given the function  $y = f(x)$  for  $1 \leq x \leq 2.5$ , where **3**

|        |     |      |     |      |      |      |     |
|--------|-----|------|-----|------|------|------|-----|
| $x$    | 1   | 1.25 | 1.5 | 1.75 | 2    | 2.25 | 2.5 |
| $f(x)$ | 3.4 | 2.2  | 0.4 | 1.9  | -2.7 | 1.3  | 2   |

Use Simpsons' rule to evaluate  $\int_1^{2.5} f(x) dx$ , correct to 1 decimal place, using the 7 function values in the table.

(c) A function is defined by  $y = g(x)$ , where  $g(x) = x^3 - 6x^2 + 5 = (x-1)(x^2 - 5x - 5).$

(i) Determine the coordinates of the stationary points and determine their nature. **3**

(ii) State at what point does the curve change its concavity? **1**

(iii) Hence sketch the graph of  $y = g(x)$ , showing all essential details. **2**

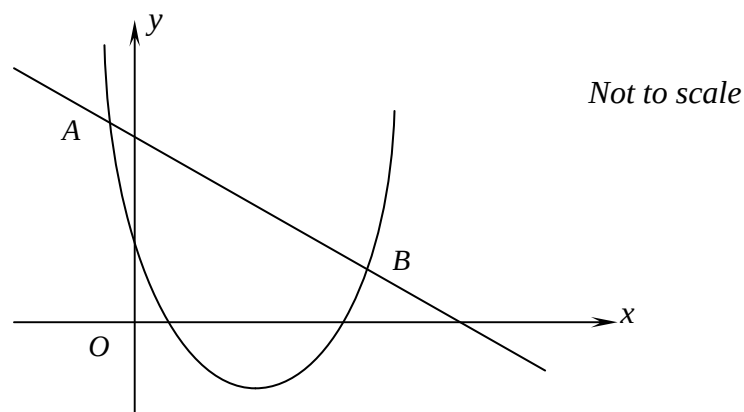
(iv) Find the minimum value of  $x^3 - 6x^2 + 5$  when  $-3 \leq x \leq 5$ . **1**

**Question 7.**

**[START A NEW PAGE]**

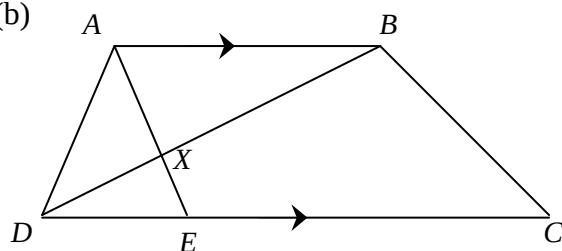
**Marks**

- (a) The parabola  $y = (x - 3)^2 - 1$  and the line  $y = 13 - 2x$  intersect at points A and B.



- (i) Find the  $x$ -coordinate for points A and B. 2
- (ii) Hence, or otherwise find the area bounded by the parabola and the line. 2

(b)



$ABCD$  is a trapezium where  $AB \parallel DC$ .  
Also  $BD \perp BC$  and  $AE \perp BD$  at  $X$ .

- (i) **Copy** the diagram into your writing booklet and find  $AX$ ,  
given  $AD = 41$  cm,  $DX = 9$  cm. 1
- (ii) What type of quadrilateral is  $ABCE$ ? Give reasons. 2
- (iii) Show that  $\triangle DXE \parallel \triangle DBC$ . 2
- (iv) Hence show that  $BX \cdot XE = 360$ . 3

**Question 8.**

**[START A NEW PAGE]**

**Marks**

- (a) A function is defined by the following properties:

**3**

$$y = 0 \text{ when } x = 1; \quad \frac{dy}{dx} = 0 \text{ when } x = -3, 1 \text{ and } 5;$$

$$\text{and } \frac{d^2y}{dx^2} > 0 \text{ for } x < -1 \text{ and } 1 < x < 3.$$

Sketch a possible graph of the function.

- (b) Larsen begins his retirement with \$500 000 at the beginning of 2009. The annual interest rate is 8% *p.a.* Interest is calculated annually on the balance at the beginning of the year and is added to the remaining balance. Larsen plans to withdraw \$56 000 annually, with the first withdrawal at the end of 2009.

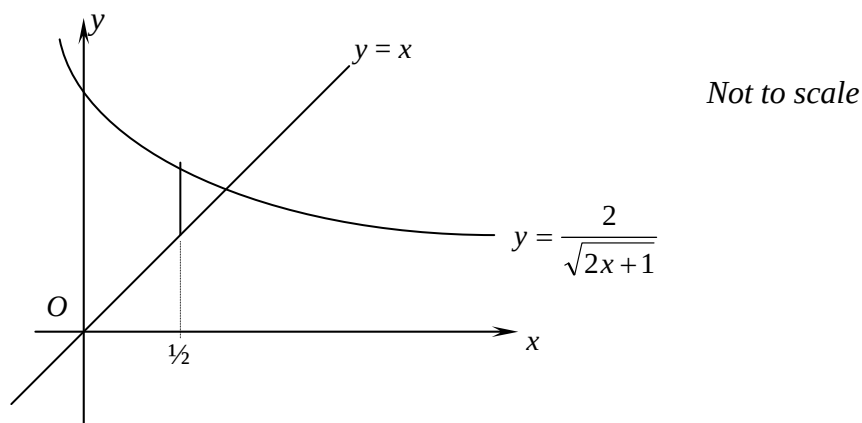
By letting  $A_n$  be the remaining balance after the  $n$ th withdrawal,

(i) Show that:  $A_2 = 5 \times 10^5 R^2 - 5 \cdot 6 \times 10^4 [1 + R]$ , where  $R = 1 \cdot 08$ . **2**

(ii) Hence deduce that:  $A_n = 10^5 [7 - 2R^n]$  **2**

(iii) Calculate during which year will Larsen's fund reach zero? **2**

- (c)



The area bounded by the curves  $y = \frac{2}{\sqrt{2x+1}}$ ,  $y = x$ , the lines  $x = 0$  and  $x = \frac{1}{2}$  **3**

is rotated about the  $x$ -axis.

Find the volume of the solid of revolution formed.

|                    |                           |              |
|--------------------|---------------------------|--------------|
| <b>Question 9.</b> | <b>[START A NEW PAGE]</b> | <b>Marks</b> |
|--------------------|---------------------------|--------------|

|     |                                         |   |
|-----|-----------------------------------------|---|
| (a) | Solve for $x$ : $x^2 - 2 x  - 15 = 0$ . | 2 |
|-----|-----------------------------------------|---|

|     |                                                                                    |   |
|-----|------------------------------------------------------------------------------------|---|
| (b) | If $\log_a(xy^3) = 1$ and $\log_a(x^2y) = 1$ . What is the value of $\log_a(xy)$ ? | 2 |
|-----|------------------------------------------------------------------------------------|---|

|     |                                                                                                                                                                                                                   |  |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| (c) | The rate at which carbon dioxide ( $\text{CO}_2$ ) will be produced when conducting an experiment is given by $\frac{dV}{dt} = 0.01(30t - t^2)$ , where $V \text{ cm}^3$ is the volume of the gas at $t$ minutes. |  |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

|     |                                                                                |   |
|-----|--------------------------------------------------------------------------------|---|
| (i) | At what rate is the gas being produced 15 minutes after the experiment begins? | 1 |
|-----|--------------------------------------------------------------------------------|---|

|      |                                                             |   |
|------|-------------------------------------------------------------|---|
| (ii) | How much carbon dioxide has been produced during this time? | 2 |
|------|-------------------------------------------------------------|---|

|     |                                                                                                                       |  |
|-----|-----------------------------------------------------------------------------------------------------------------------|--|
| (d) | The mass $S$ grams of a substance decays at an instantaneous rate proportional to its mass present at time $t$ years. |  |
|-----|-----------------------------------------------------------------------------------------------------------------------|--|

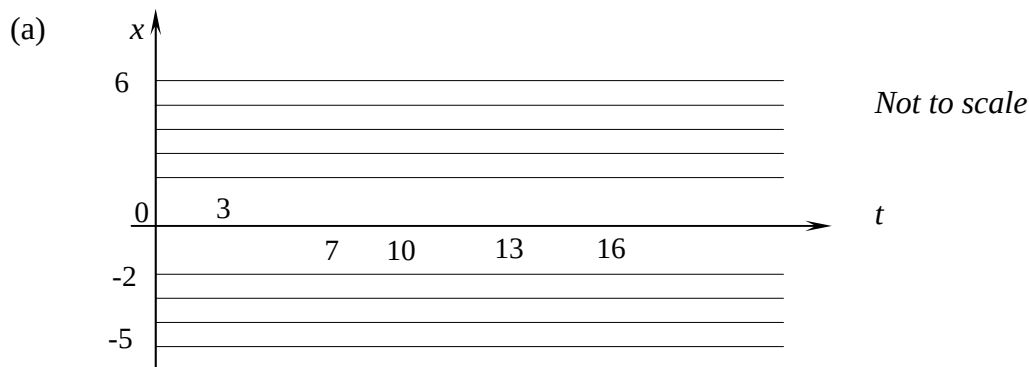
|     |                                                                                                                                      |   |
|-----|--------------------------------------------------------------------------------------------------------------------------------------|---|
| (i) | Explain why the rate equation is given by: $\frac{dS}{dt} = -kS$ ,<br>where $k (> 0)$ is the decay rate constant of proportionality. | 1 |
|-----|--------------------------------------------------------------------------------------------------------------------------------------|---|

|      |                                                                                                                               |   |
|------|-------------------------------------------------------------------------------------------------------------------------------|---|
| (ii) | Verify that $S = S_0 e^{-kt}$ is the general solution to the rate equation, where $S_0$ is the initial mass of the substance. | 1 |
|------|-------------------------------------------------------------------------------------------------------------------------------|---|

|       |                                                                               |   |
|-------|-------------------------------------------------------------------------------|---|
| (iii) | Show that the half-life for the substance is given by $\frac{\ln 2}{k}$ years | 1 |
|-------|-------------------------------------------------------------------------------|---|

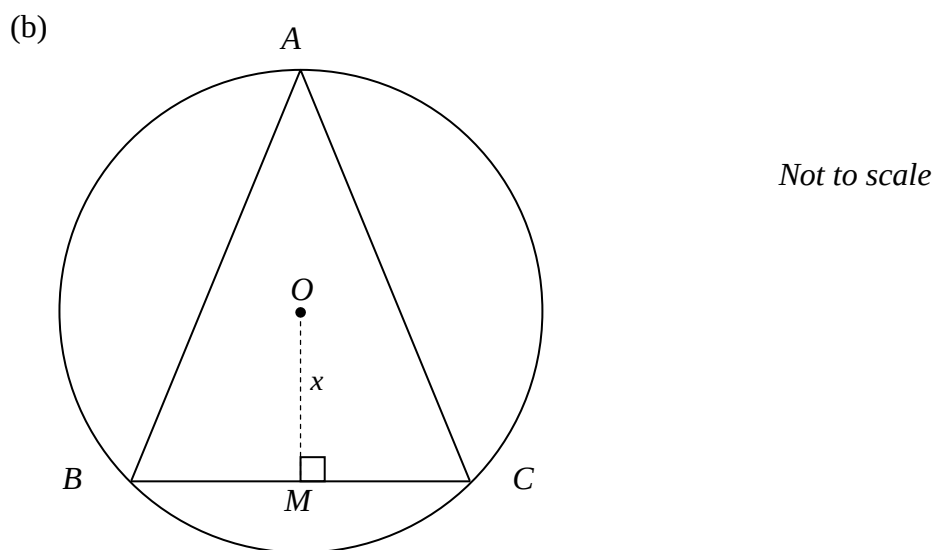
|      |                                                                                            |   |
|------|--------------------------------------------------------------------------------------------|---|
| (iv) | If the mass present at time $t_1$ is $S_1$ and the mass at time $t_2$ is $S_2$ , show that | 2 |
|------|--------------------------------------------------------------------------------------------|---|

$$k = \frac{1}{t_2 - t_1} \ln \left( \frac{S_1}{S_2} \right), \text{ assuming } t_2 > t_1.$$



A particle is moving along a straight line according to the sketch of displacement  $x$  metres against time  $t$  seconds above.

- |       |                                                                     |   |
|-------|---------------------------------------------------------------------|---|
| (i)   | When is the particle at rest?                                       | 1 |
| (ii)  | At what time does the particle have greatest speed (approximately)? | 1 |
| (iii) | Describe what happens to the particle as $t \rightarrow \infty$ .   | 1 |
| (iv)  | Calculate the distance that the particle has eventually travelled.  | 2 |



An isosceles triangle  $ABC$  with  $AB = AC$  is inscribed in a circle centre  $O$  and of radius  $R$  units.

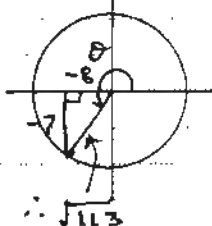
Given that  $OM = x$  units,  $OM \perp BC$  and  $M$  is the midpoint of  $BC$ ,

- |      |                                                                            |   |
|------|----------------------------------------------------------------------------|---|
| (i)  | Show that the area of $\triangle ABC$ , $S$ square units, is given by:     | 2 |
|      | $S = (R + x)\sqrt{R^2 - x^2}.$                                             |   |
| (ii) | Hence show that the triangle with maximum area is an equilateral triangle. | 5 |

# 20 MATHEMATICS TRIAL, 2009.

## MARKING SCHEME

### MATHEMATICS: Question 1

| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks | Marker's Comments |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|--|-----|
| <p>1(a) <math>\frac{2^3 - 2 \cdot 1^2}{\sqrt{3 \cdot 14 + 2 \cdot 1}} = \frac{2^3 - 4 \cdot 1}{\sqrt{5 \cdot 24}} = \frac{6 \cdot 8478901...}{\sqrt{120}} = 6.8 \quad (2 \text{ SF}) \checkmark</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                            |       | [1]               |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |     |
| <p>(b) <math>\tan \theta = \frac{1}{8} \quad \cos \theta &lt; 0 \Rightarrow 3^{\text{rd}} \text{ Quadrant}</math></p>  <p><math>\cos \theta = \frac{1}{\sin \theta} = -\frac{\sqrt{113}}{7}</math></p>                                                                                                                                                                                                                                                                                                                                                                |       | [2]               |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |     |
| <p>(c) <math> 4x - 15  \leq 3</math></p> <p><math>\therefore -3 \leq 4x - 15 \leq 3</math></p> <p><math>12 \leq 4x \leq 18</math></p> <p><math>3 \leq x \leq 9\frac{1}{2} \quad \checkmark</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       | [2]               |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |     |
| <p>(d) <math>3y = \sin(2x - \frac{\pi}{4})</math></p> <p><math>\therefore y = \frac{1}{3} \sin(2x - \frac{\pi}{4})</math></p> <p>Period = <math>\frac{2\pi}{2} = \pi</math>     Amplitude = <math>\frac{1}{3}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                              |       | [2]               |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |     |
| <p>(e) SELLING = <math>130\% \text{ cost} = \\$67.60</math></p> <p><math>\therefore \text{cost} = \frac{\\$67.60 \times 100}{130}</math></p> <p><math>= \\$52 \quad \checkmark</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       | [2]               |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |     |
| <p>(f) <math>\tan \alpha = -0.5 \quad 0 &lt; \alpha &lt; \pi \quad 2^{\text{nd}} \text{ Q.}</math></p> <p><math>\alpha = \pi - 0.4676...</math></p> <p><math>= 2.677945...</math></p> <p><math>\alpha = 2.68 \quad (2 \text{ dp}) \quad \checkmark</math></p>                                                                                                                                                                                                                                                                                                                                                                                         |       | [1]               |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |     |
| <p>(g)</p> <table style="margin-left: auto; margin-right: auto;"> <tr><td>11</td><td>21</td><td>31</td><td>41</td><td>51</td><td>61</td></tr> <tr><td>12</td><td>22</td><td>32</td><td>42</td><td>52</td><td>62</td></tr> <tr><td>13</td><td>23</td><td>33</td><td>43</td><td>53</td><td>63</td></tr> <tr><td>14</td><td>24</td><td>34</td><td>44</td><td>54</td><td>64</td></tr> <tr><td>15</td><td>25</td><td>35</td><td>45</td><td>55</td><td>65</td></tr> <tr><td>16</td><td>26</td><td>36</td><td>46</td><td>56</td><td>66</td></tr> </table> <p><math>P(E = \text{perfect eq. digits}) = \frac{8}{36} = \frac{2}{9} \quad \checkmark</math></p> | 11    | 21                | 31 | 41 | 51 | 61 | 12 | 22 | 32 | 42 | 52 | 62 | 13 | 23 | 33 | 43 | 53 | 63 | 14 | 24 | 34 | 44 | 54 | 64 | 15 | 25 | 35 | 45 | 55 | 65 | 16 | 26 | 36 | 46 | 56 | 66 |  | [2] |
| 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 21    | 31                | 41 | 51 | 61 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |     |
| 12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 22    | 32                | 42 | 52 | 62 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |     |
| 13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 23    | 33                | 43 | 53 | 63 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |     |
| 14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 24    | 34                | 44 | 54 | 64 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |     |
| 15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 25    | 35                | 45 | 55 | 65 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |     |
| 16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 26    | 36                | 46 | 56 | 66 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |     |

# MATHEMATICS: Question 2

| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                           | Marks | Marker's Comments |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------|
| <p>Q 2(a) <math>m_{AC} = \frac{4\frac{1}{2} - 0}{0 - 5} = -\frac{4\frac{1}{2}}{5} = -\frac{9}{10}</math> ✓</p> <p>Eqn. of AC : <math>y = mx + b</math></p> <p><math>y = -\frac{9}{10}x + 4\frac{1}{2}</math> ✓</p> <p><math>10y = -9x + 45</math></p> <p><math>\therefore 9x + 10y - 45 = 0</math> qed</p>                                                                                                                    |       | 2                 |
| <p>b) <math>AC^2 = 5^2 + 4\frac{1}{2}^2</math> (Pyth Thm)</p> <p><math>AC^2 = \frac{181}{4}</math></p> <p><math>\therefore AC = \frac{\sqrt{181}}{2}</math> units ✓</p>                                                                                                                                                                                                                                                       |       | 1                 |
| <p>(c) <math>x - 2y + 9 = 0</math> — (1) : AB</p> <p><math>4x - y - 20 = 0</math> — (2) : CB</p> <p><math>\therefore \frac{8x - 2y - 40 = 0}{7x - 4y = 0}</math> — (2a)</p> <p><math>\therefore x = 7</math> ✓</p> <p>Subst in (2) <math>28 - y - 20 = 0</math></p> <p><math>\therefore y = 8</math> ✓</p> <p><math>\therefore B = (7, 8)</math></p>                                                                          |       | 2                 |
| <p>(d) <math>m_{AB} = \frac{1}{2}</math> ✓ from <math>x - 2y + 9 = 0</math></p> <p><math>\tan \theta = \frac{1}{2}</math> <math>\angle \theta = 26^\circ 34' \div 27^\circ</math> ✓</p>                                                                                                                                                                                                                                       |       | 2                 |
| <p>(e) AC : <math>9x + 10y - 45 = 0</math></p> <p>B : (7, 8)</p> <p><math>\perp</math> dist = <math>\frac{ 9 \times 7 + 10 \times 8 - 45 }{\sqrt{9^2 + 10^2}} = \frac{98}{\sqrt{181}}</math> ✓</p> <p><math>\therefore \text{AREA } \triangle ABC = \frac{1}{2} \times AC \times \perp d.</math></p> <p><math>= \frac{1}{2} \times \frac{\sqrt{181}}{2} \times \frac{98}{\sqrt{181}} = 24\frac{1}{2} \text{ u}^2</math> ✓</p> |       | 2                 |
| <p>(f)</p> <p>AC : <math>9x + 10y - 45 \geq 0</math></p> <p>BC : <math>4x - y - 20 \leq 0</math></p> <p>AB : <math>x - 2y + 9 \geq 0</math></p>                                                                                                                                                                                                                                                                               |       | 3                 |

# MATHEMATICS: Question 3

| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks | Marker's Comments                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------|
| <p>Q3(a) (i) <math>y = (1 + \ln x)^2</math><br/> <math>\frac{dy}{dx} = 2(1 + \ln x) \times \frac{1}{x} = \frac{2(1 + \ln x)}{x}</math> ✓</p> <p>(ii) <math>y = \frac{\sin x}{e^{3x} + 1}</math> : <math>\frac{dy}{dx} = \frac{\cos x(e^{3x} + 1) - \sin x \times 3e^{3x}}{(e^{3x} + 1)^2}</math> ✓✓<br/> <math>\frac{dy}{dx} = \frac{(e^{3x} + 1) \cos x - 3 \sin x \cdot e^{3x}}{(e^{3x} + 1)^2}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       | <p>[2]</p> <p>[2]</p>            |
| <p>(b) (i) <math>\int (5x - 1)^3 dx = \frac{1}{4 \times 5} (5x - 1)^4 + C</math> ✓✓<br/> <math>= \frac{1}{20} (5x - 1)^4 + C</math></p> <p>(ii) <math>\int_0^{\pi} \sec^2 \frac{x}{4} dx = 4 \left[ \tan \frac{x}{4} \right]_0^{\pi}</math> ✓<br/> <math>= 4 \left[ \tan \frac{\pi}{4} - \tan 0 \right]</math><br/> <math>= 4 [1 - 0] = 4</math> ✓</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       | <p>[2]</p> <p>[2]</p>            |
| <p>(c) (i) <math>2x^2 - 6x - 7 = 0</math><br/> <math>\therefore \alpha + \beta = -\frac{b}{a} = -\frac{(-6)}{2} = 3</math> ✓</p> <p>(ii) <math>\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta</math> ✓<br/> <math>= 3^2 - 2 \times \left( -\frac{7}{2} \right) = 9 + 7</math><br/> <math>= 16</math> ✓</p> <p>or <math>2\alpha^2 - 6\alpha - 7 = 0</math> as <math>\alpha</math> is a root<br/> <math>2\beta^2 - 6\beta - 7 = 0</math><br/> <math>2(\alpha^2 + \beta^2) - 6(\alpha + \beta) - 14 = 0</math><br/> <math>2(\alpha^2 + \beta^2) - 6 \times 3 - 14 = 0</math><br/> <math>\therefore \alpha^2 + \beta^2 = \frac{32}{2} = 16</math></p> <p>(iii) As <math>\alpha</math> is a root<br/> <math>2\alpha^2 - 6\alpha - 7 = 0</math><br/> <math>2\alpha^2 - 6\alpha = 7</math><br/> <math>\therefore \alpha^2 - 3\alpha = \frac{7}{2} = 3\frac{1}{2}</math> ✓</p> |       | <p>[1]</p> <p>[2]</p> <p>[1]</p> |

# MATHEMATICS: Question 4

| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks                                                                                                                         | Marker's Comments |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <p>Q4(a)(i)</p> <p> <math>\angle ABC = 120^\circ</math><br/> <math>\angle BAC = 210^\circ</math><br/> <math>\angle ABC = 120^\circ</math> </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                               | 1                 |
| <p>(ii)</p> $AC^2 = 50^2 + 60^2 - 2 \times 50 \times 60 \times \cos 120^\circ$ $= 9100$ $AC = 10\sqrt{91} = 95.39392014...$ <p><math>\therefore AC = 95</math> n.miles (nearest nm) ✓</p>                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                               | 2                 |
| <p>(iii)</p> <p>Let <math>\angle CAB = \theta^\circ</math></p> $\frac{\sin \theta}{50} = \frac{\sin 120^\circ}{AC}$ $\sin \theta = \frac{50 \times \sin 120^\circ}{AC} = \begin{cases} 0.4539206... \\ 0.455802... \end{cases}$ $\angle \theta = \begin{cases} 27^\circ \text{ or } 153^\circ \\ 27^\circ 7' \text{ or } 152^\circ 53' \end{cases} \text{ but } 120^\circ + 152^\circ > 180^\circ$ <p><math>\therefore \angle \theta = 27^\circ \text{ or } 27^\circ 7'</math> ✓</p> <p><math>\therefore</math> Bearing of C from A is <math>243^\circ \text{ T} / 242^\circ 53' \text{ T}</math> ✓</p> |                                                                                                                               | 2                 |
| <p>(b)</p> $\lim_{x \rightarrow 16} \frac{x-16}{\sqrt{x}-4} = \lim_{x \rightarrow 16} \frac{(\sqrt{x}-4)(\sqrt{x}+4)}{\sqrt{x}-4}$ $= \lim_{x \rightarrow 16} (\sqrt{x}+4) \text{ as } x \neq 16$ $= \sqrt{16}+4 = 8$ ✓                                                                                                                                                                                                                                                                                                                                                                                 | <p>(or)</p> $\frac{x-16}{\sqrt{x}-4} \times \frac{\sqrt{x}+4}{\sqrt{x}+4}$ $= \frac{(x-16)(\sqrt{x}+4)}{x-16}$ $= \sqrt{x}+4$ | 2                 |
| <p>(c)</p> $y = 1 + \ln 2x$ $\frac{dy}{dx} = \frac{1}{2x} \times 2 = \frac{1}{x}$ <p>Gradient of Tangent at <math>x = \frac{1}{2}e</math>: <math>m_T = \frac{2}{e}</math></p> <p>" of Normal <math>m_N = -\frac{e}{2}</math> ✓</p> <p>Eqn. of Normal at <math>(\frac{1}{2}e, 2)</math>: <math>y-2 = -\frac{e}{2}(x-\frac{1}{2}e)</math> ✓</p> <p><math>\therefore y = -\frac{1}{2}ex + \frac{1}{4}e^2 + 2</math></p>                                                                                                                                                                                    | $2ex + 4y - e^2 - 8 = 0$                                                                                                      | 3                 |
| <p>(d)</p> $4x^2 - 4x + 4y^2 + 24y + 21 = 0$ $x^2 - x + \frac{1}{4} + y^2 + 6y + 9 = -\frac{21}{4} + \frac{1}{4} + 9$ $(x - \frac{1}{2})^2 + (y + 3)^2 = 4$ ✓                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                               | 2                 |
| <p><math>\therefore</math> centre is <math>(\frac{1}{2}, -3)</math> ✓ radius = 2</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                               | 2                 |



# MATHEMATICS: Question 6

6

## Suggested Solutions

Marks

Marker's Comments

26(a) LHS =  $\frac{1 + \tan^2 A}{\cos^2 A} = \frac{\sec^2 A}{\cos^2 A}$  ✓

Approach I:

$$\begin{aligned} &= \frac{1}{\cos^2 A} \div \frac{1}{\sin^2 A} \\ &= \frac{\sin^2 A}{\cos^2 A} \quad \checkmark \\ &= \tan^2 A \quad \text{400.} \end{aligned}$$

II:  $\frac{1 + \tan^2 A}{\cos^2 A} = \frac{\sin^2 A (1 + \frac{1}{\sin^2 A})}{\cos^2 A}$   
 $= \frac{\sin^2}{\cos^2} (1 + \frac{1}{\sin^2})$   
 $= \frac{\sin^2}{\cos^2} = \tan^2 A$

2

(b)

|      |     |      |     |      |      |      |     |
|------|-----|------|-----|------|------|------|-----|
| x    | 1   | 1.25 | 1.5 | 1.75 | 2    | 2.25 | 2.5 |
| f(x) | 3.4 | 2.2  | 0.4 | 1.9  | -2.7 | 1.3  | 2   |

$y_1 \quad y_2 \quad y_3 \quad y_4 \quad y_5 \quad y_6 \quad y_7$

$h = \frac{2.5 - 1}{6} = \frac{1.5}{6} = \frac{1}{4} = 0.25$  ✓

$$\begin{aligned} \int f(x) dx &\approx A_3 \approx \frac{h}{3} [y_1 + y_7 + 4(y_2 + y_4 + y_6) + 2(y_3 + y_5)] \\ &= \frac{0.25}{3} [5.4 + 4(2.2 + 1.9 + 1.3) + 2(0.4 + -2.7)] \\ &= \frac{0.25}{3} [5.4 + 4 \times 5.4 + 2 \times (-2.3)] \\ &= \frac{0.25}{3} [5.4 + 21.6 - 4.6] = \frac{0.25}{3} \times 22.4 \\ &= 1.86 \\ &= 1.9 \quad (1 \text{ dp}) \end{aligned}$$

3

(c)  $g(x) = x^3 - 6x^2 + 5 = (x-1)(x^2 - 5x - 5)$

(i)  $g'(x) = 3x^2 - 12x = 3x(x-4)$

$g''(x) = 6x - 12 = 6(x-2)$

For SPs to occur  $g'(x) = 0$

i.e.  $3x(x-4) = 0$

$x = 0$  or  $4$

$\therefore g(0) = 5$  or  $-27$

$\therefore$  SPs are  $(0, 5)$  and  $(4, -27)$

TEST: at  $x=0$

$g''(0) = -12 < 0$

$\therefore$  concave downwards

$\therefore$  Rel. max TP at  $(0, 5)$

at  $x=4$

$g''(4) = 12 > 0$

$\therefore$  concave upwards

$\therefore$  Rel. min TP at  $(4, -27)$

3

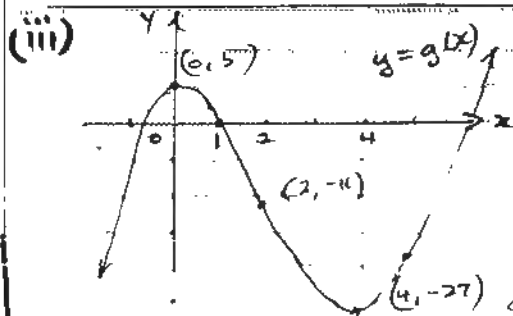
(ii) change in concavity can be at  $g''(x) = 0$  P.O.I.

so at  $x=2$

$\therefore (2, -11)$  ✓

1

|        |    |   |   |
|--------|----|---|---|
| x      | 1  | 2 | 3 |
| g''(x) | -6 | 0 | 6 |



2

(iv)  $-3 \leq x \leq 5$

$g(-3) = -76$

$g(5) = -20$

$\therefore$  From sketch  $y = g(x)$

Absolute min is  $-76$  at  $x = -3$  ✓

1

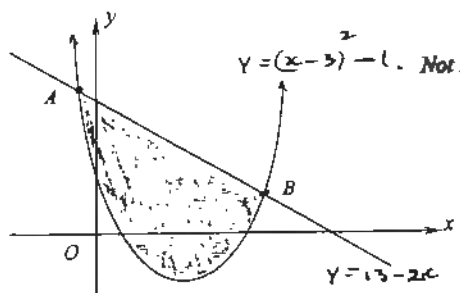
# MATHEMATICS: Question 7.

Suggested Solutions

Marks

Marker's Comments

Q7(a)  $y = (x-3)^2 - 1$   
 $y = 13 - 2x$



(i) POINTS OF INTERSECTION

$$(x-3)^2 - 1 = 13 - 2x$$

$$x^2 - 6x + 9 - 1 = 13 - 2x$$

$$x^2 - 4x - 5 = 0$$

$$(x+1)(x-5) = 0$$

$$\therefore x = -1 \text{ or } 5$$

$$x_A = -1 \quad x_B = 5$$

[2]

(ii) Area =  $\int_{-1}^5 (y_U - y_L) dx$

$$= \int_{-1}^5 (13 - 2x - \{(x-3)^2 - 1\}) dx$$

$$= \int_{-1}^5 (14 - 2x - (x-3)^2) dx = \int_{-1}^5 (5 + 4x - x^2) dx$$

$$= \left[ 5x - x^2 - \frac{1}{3}(x-3)^3 \right]_{-1}^5 = \left[ 5x + 2x^2 - \frac{1}{3}x^3 \right]_{-1}^5$$

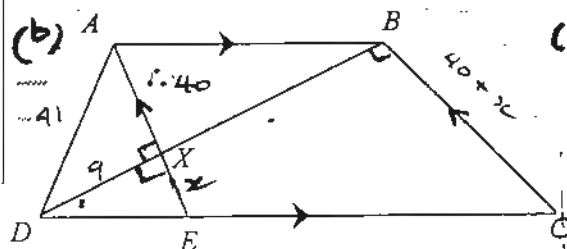
$$= \left[ (70 - 25 - \frac{8}{3}) - (-14 - 1 + \frac{64}{3}) \right] = 78 - 42$$

$$= 42\frac{1}{3} - 6\frac{1}{3}$$

$$\text{Area} = 36 \text{ sq. units}$$

$$= 36$$

[2]



(i)  $41^2 = 9^2 + Ax^2$  (Pyth. Thm.)

$$Ax = 40 \text{ cm}$$

[1]

(ii) As  $\angle DXE = \angle DBC = 90^\circ$  and are corresponding angles ✓  
 $\therefore AE \parallel CB$  and as  $AB \parallel DC$   
 $\therefore ABCE$  is a parallelogram ✓ (two pair of parallel sides)

[2]

(iii) In  $\Delta s DXE$  and  $DBC$

1.  $\angle XDE = \angle BDC$  (common) ✓

2.  $\angle DXE = \angle DBC = 90^\circ$  (data)

$\therefore \Delta DXE \parallel \Delta DBC$  (equiangular / matching angles equal) ✓

[2]

(iv)  $AE = 40 + xE$

but  $BC = AE$  (opp. sides of ||ogram are equal) ✓

$\therefore \frac{BC}{XE} = \frac{DB}{DX}$  (corresponding sides in similar triangles are in the same ratio)

or  $\frac{XE}{BC} = \frac{DX}{DB}$

$$\frac{XE}{40+xE} = \frac{9}{9+BX}$$

i.e.  $\frac{40+xE}{XE} = \frac{9+BX}{9}$  ✓

$$360 + 9XE = 9XE + XB \cdot XE$$

$$\therefore BX \cdot XE = 360 \text{ qed.}$$

[3]

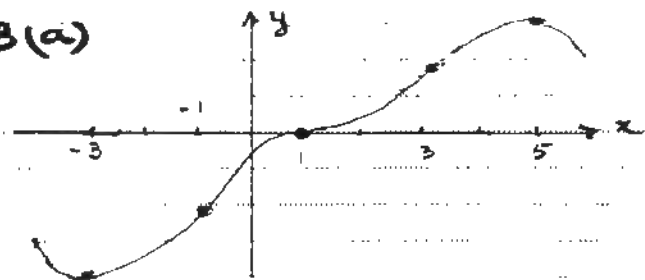
# MATHEMATICS: Question 8

## Suggested Solutions

Marks

Marker's Comments

Q8(a)



Possible sketch.

Rel min TP  $x = -3$  ✓  
 Rel max TP  $x = 5$  ✓  
 P.O.I. at  $x = -1, 3$  ✓  
 HPOI at  $(1, 0)$  ✓

3

b)  $r = 8\% = 0.08$   
 $A = 1 + \frac{r}{100} = 1.08$  given

(i) Begin 2010  $A_1 = 500000 \times A - 56000$  ✓  
 $=$   
 Begin 2011  $A_2 = A_1 A - 56000$   
 $= (5 \times 10^5 A - 5.6 \times 10^4) A - 5.6 \times 10^4$  ✓  
 $= 5 \times 10^5 A^2 - 5.6 \times 10^4 (1 + A)$  qed

2

(ii) so Begin 2012  $A_3 = 5 \times 10^5 A^3 - 5.6 \times 10^4 [1 + A + A^2]$   
 $\therefore$  After  $n$  years  
 $A_n = 5 \times 10^5 A^n - 5.6 \times 10^4 [1 + A + A^2 + \dots + A^{n-1}]$  ✓  
 $= 5 \times 10^5 A^n - 5.6 \times 10^4 \left[ \frac{A^n - 1}{A - 1} \right]$  ✓  
 $= 5 \times 10^5 A^n - 5.6 \times 10^4 \left[ \frac{A^n - 1}{0.08} \right]$  ;  $A - 1 = 1.08 - 1$   
 $= 5 \times 10^5 A^n - 7 \times 10^5 (A^n - 1)$  ✓  
 $= 5 \times 10^5 A^n - 7 \times 10^5 A^n + 7 \times 10^5$  ✓  
 $\therefore A_n = 10^5 [7 - 2A^n]$  qed

2

(iii) when  $A_n = 0$   
 $\therefore 7 - 2A^n = 0$   
 $1.08^n = \frac{7}{2} = 3.5$  ✓  
 $n = \frac{\ln 3.5}{\ln 1.08} = 16.27788 \dots$

$\therefore$  Funds runs out during 17<sup>th</sup> year / (but actually at end of) 2025 ✓

2

(c) Vol  $= \pi \int_{-1}^2 y_U^2 dx - \pi \int_{-1}^2 y_L^2 dx$   
 $= \pi \int_{-1}^2 \left( \frac{4}{2x+1} - x^2 \right) dx$  ✓  
 $= \pi \left[ 2 \ln(2x+1) - \frac{1}{3} x^3 \right]_0^2$  ✓  
 $= \pi \left[ 2 \ln 5 - \frac{8}{3} - 0 \right]$

Vol  $= \pi \left( \ln 4 - \frac{1}{3} \right)$  units<sup>3</sup> ✓

3



## MATHEMATICS

: Question 9 CONT.

## Suggested Solutions

Marks

Marker's Comments

Q9(d)(i) Rate of decay is proportional to mass present (data) ✓

$$\therefore \frac{dS}{dt} \propto -S$$

$$\therefore \frac{dS}{dt} = -kS, \text{ where } k \text{ is } \dots$$

11

$$(ii) S = S_0 e^{-kt}$$

$$\text{LHS: } \frac{dS}{dt} = S_0 \times -k e^{-kt}$$

$$= -k S_0 e^{-kt}$$

$$\therefore \text{LHS} = \text{RHS}$$

 $\therefore S = S_0 e^{-kt}$  satisfies rate equation.

11

(iii)

$$\text{When } S = \frac{1}{2} S_0$$

$$\text{as } t=0 \quad S=S_0$$

$$\therefore \frac{1}{2} S_0 = S_0 e^{-kt}$$

$$e^{-kt} = \frac{1}{2}$$

$$-kt = \ln \frac{1}{2} = -\ln 2 \quad \checkmark$$

$$t_{\frac{1}{2}} = \frac{\ln 2}{k} \quad \text{qed.}$$

$$e^{kt} = \frac{2}{1}$$

$$kt = \ln 2$$

$$t = \frac{\ln 2}{k}$$

11

(iv)

$$E_1: S_1 = S_0 e^{-kt_1} \quad \text{--- (1)}$$

$$E_2: S_2 = S_0 e^{-kt_2} \quad \text{--- (2)} \quad t_2 > t_1$$

$$\therefore \frac{S_1}{S_2} = e^{-kt_1 + kt_2} = e^{-k(t_1 - t_2)} \quad \checkmark$$

$$\ln \left( \frac{S_1}{S_2} \right) = -k(t_1 - t_2) = k(t_2 - t_1) \quad \checkmark$$

$$\therefore k = \frac{1}{t_2 - t_1} \ln \left( \frac{S_1}{S_2} \right) \quad \text{qed.}$$

2

(v) -

**MATHEMATICS: Question 10**

**Suggested Solutions**

**Marks**

**Marker's Comments**

Q10(a).

(i) At rest when  $v = \frac{dx}{dt} = 0$

$\therefore$  at  $t = 3$  and  $10$  (3 and 10 seconds)

1

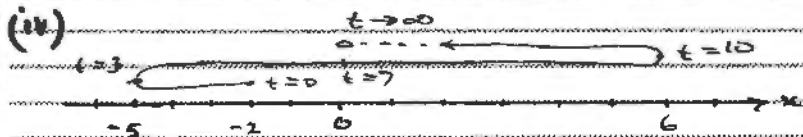
(ii) At greatest speed  $\frac{dv}{dt} = 0$  and  $\frac{dx}{dt} = \tan \theta$

$\therefore$  at approx 7 seconds ( $t = 7$ )

1

(iii) the particle approaches 0 from  $x > 0$   
at ever decreasing speed ( $v < 0$  and  $\ddot{x} > 0$ )

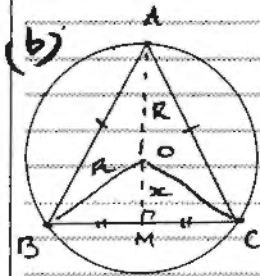
1



Total distance =  $3 + \dots + 6$   
 $= 20 \text{ m}$

$-2 \rightarrow -5 : 3 \text{ m}$   
 $-5 \rightarrow 6 : 11 \text{ m}$   
 $6 \rightarrow 0 : 6 \text{ m}$

2



(i) Area of  $\triangle ABC$   $S = \frac{1}{2} BC \times h = \frac{1}{2} \times BC \times AM$ .

$$R^2 = x^2 + BM^2 \quad (\text{Pyth. thm})$$

$$BM^2 = R^2 - x^2$$

$$BM = \sqrt{R^2 - x^2}$$

$$\therefore BC = 2BM = 2\sqrt{R^2 - x^2}$$

$$h = AM = x + R$$

$$\therefore S = \frac{1}{2} \times 2\sqrt{R^2 - x^2} \times (R + x)$$

$$\text{i.e. } S = (R + x)\sqrt{R^2 - x^2} \quad 0 \leq x \leq R$$

2

(ii)  $\frac{dS}{dx} = 1 \cdot \sqrt{R^2 - x^2} + (R + x) \cdot \frac{1}{2} (R^2 - x^2)^{-\frac{1}{2}} \cdot (-2x)$

$$= \sqrt{R^2 - x^2} - \frac{x(R + x)}{\sqrt{R^2 - x^2}}$$

$$= \frac{R^2 - x^2 - Rx - x^2}{\sqrt{R^2 - x^2}}$$

$$\frac{dS}{dx} = \frac{R^2 - Rx - 2x^2}{\sqrt{R^2 - x^2}}$$

For possible max/min values of  $S$  to occur  $\frac{dS}{dx} = 0$

$$\text{So } R^2 - Rx - 2x^2 = 0 \quad \text{only}$$

$$2x^2 + Rx - R^2 = 0$$

$$(2x - R)(x + R) = 0$$

$$x = -R \text{ or } \frac{1}{2}R$$

$$\text{So } x = \frac{1}{2}R \quad \checkmark \text{ as } x > 0 \text{ and } 0 \leq x \leq R$$

$$\therefore S = \frac{3R}{2} \sqrt{R^2 - R^2} = \frac{3\sqrt{2}}{4} R^2$$

continued  $\rightarrow$

# MATHEMATICS: Question 10 CONT.

## Suggested Solutions

## Marks

## Marker's Comments

TEST nature at  $x = \frac{1}{2}R$

METHOD 1

|                                                 |                 |       |                |                                                 |
|-------------------------------------------------|-----------------|-------|----------------|-------------------------------------------------|
| $R^2 - Rx - 2x^2 =$                             | $x$             | $0$   | $\frac{1}{2}R$ | $\frac{1}{2}R$ or ...                           |
| $\frac{dS}{dx} = \frac{d}{dx}(R^2 - Rx - 2x^2)$ | $\frac{dS}{dx}$ | $R$   | $0$            | $-R^2/2$ (-)                                    |
| $\frac{dS}{dx} = \frac{d}{dx}(R^2 - Rx - 2x^2)$ |                 | $(+)$ |                | $\frac{dS}{dx} = \frac{d}{dx}(R^2 - Rx - 2x^2)$ |

$\therefore$  A Rel max TP at  $x = \frac{1}{2}R$  ✓

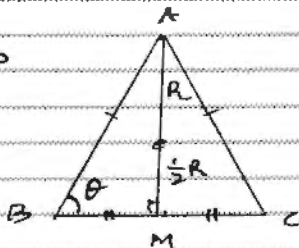
METHOD 2:  $x=0$   $S=R^2$

$$x=R \quad S = \frac{3\sqrt{2}R^2}{4} > R^2$$

$$x=R \quad S=0$$

and since  $S$  is continuous for  $0 \leq x \leq R$   
so Rel max TP is the absolute max at  $x = \frac{1}{2}R$

Now



$$BM = \sqrt{R^2 - x^2} = \sqrt{R^2 - \frac{R^2}{4}} = \frac{R\sqrt{3}}{2}$$

$$\therefore BC = R\sqrt{3}$$

$$\tan \theta = \frac{R + \frac{1}{2}R}{\frac{R\sqrt{3}}{2}} = \frac{3}{\sqrt{3}} = \sqrt{3}$$

$$\therefore \theta = 60^\circ$$

$\therefore \angle C = \angle B = 60^\circ$  (equal angles opposite equal sides)  
(data)

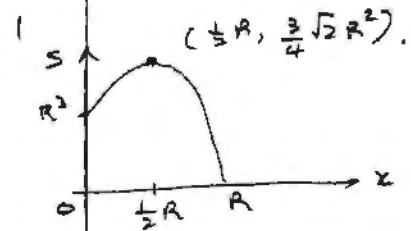
$$\therefore \angle A = 60^\circ$$

$\therefore$  max area when  $x = \frac{1}{2}R$  and  $\triangle ABC$  is equilateral.  
qed

5

$$x = 0.7R \quad \frac{dS}{dx} = -0.6 \times 1.7R^2 < 0$$

$$x = 0.9R \quad \frac{dS}{dx} = -0.2 \times 1.9R^2 < 0$$



Ans 2: For  
A proof that  $\triangle ABC$   
is equilateral  
when  $x = \frac{1}{2}R$